

1.2.4. Pattern - 2

00:28

Write a Python program to print a right-angled triangle pattern of numbers.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format:

- The output should display the pattern of numbers separated by space, with each row containing increasing numbers starting from 1 up to the row number.

Note:

- Refer to the displayed test cases for the sample pattern.

Sample Test Cases

+

Explorer

numberP...

Submit

```
1 # Type Content here...
2 def print_number_triangle(rows):
3     for i in range(1, rows + 1):
4         for j in range(1, i + 1):
5             print(j, end=" ")
6         print()
7
8 # Input: Number of rows in the pattern
9 rows = int(input())
10
11 # output: Right-angled triangle pattern of numbers
```

Average time

0.004 s

4.25 ms

Maximum time

0.005 s

5.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 5 ms

Debug

⋮

⌵

⤴

Expected output

Actual output

5

5

1

1

1 2

1 2

1 2 3

1 2 3

1 2 3 4

1 2 3 4

1 2 3 4 5

1 2 3 4 5

Terminal

Test cases

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1.2.3. Pattern - 1

00:12

Write a Python program to print a pattern of asterisks in the form of a right-angled triangle.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format

- The output should display the pattern of asterisks (*), with each row containing an increasing number of asterisks.

Note: Refer to the displayed test cases for the sample pattern.

Sample Test Cases

+

rightangl...

Submit

```
1 #Type Content here...
2 rows = int(input())
3
4 for i in range(1, rows + 1):
5     # Each row contains 'i' asterisks separated by
    spaces
6     print("* " * i)
```

Average time
0.004 s
3.83 ms

Maximum time
0.005 s
5.00 ms

2 out of 2 shown test case(s) passed
4 out of 4 hidden test case(s) passed

Test case 1 4 ms Debug

Expected output
5
*
* *
* * *
* * * *
* * * * *

Actual output
5
*
* *
* * *
* * * *
* * * * *

Terminal

Test cases

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1.2.1. Pass or Fail

1.2.2. Fibonacci Series

1.2.3. Pattern - 1

1.2.4. Pattern - 2

2. Practical 2

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1.2.2. Fibonacci Series

Write a Python program that uses recursion to print the first n terms of the Fibonacci series.

Input Format:

A single integer n representing the number of terms to generate.

Output Format:

A single line containing the first n terms of the Fibonacci sequence, separated by spaces.

Sample Test Cases

fibonacci....

```
1 '''# def fibonacci(n):
2
3 # -># write the code..
4
5 # ->return fibonacci(...)
6
7
8 # n = int(input())
9 # for i in range(1, n + 1):
10 # ->print(fibonacci(i), end=" ")
11 '''
```

Average time0.021 s20.50 ms

Maximum time0.065 s65.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 17 ms

Expected output5

Actual output5

0 1 1 2 3

Test case 24 ms

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Test cases

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1.1. Practice Lab Assignment

1.1.1. Calculate Momentum

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1.2.1. Pass or Fail 03:23

Write a Python program that accepts the number of courses and the marks of a student in those courses.
The grade is determined based on the aggregate percentage:

- If the aggregate percentage is greater than 75, the grade is Distinction.
- If the aggregate percentage is greater than or equal to 60 but less than 75, the grade is First Division.
- If the aggregate percentage is greater than or equal to 50 but less than 60, the grade is Second Division.
- If the aggregate percentage is greater than or equal to 40 but less than 50, the grade is Third Division.

Input Format:

- The first input will be an integer n , the number of courses.
- The second input will be n integers representing the marks of the student in each of the n courses, separated by a space.

Output Format:

If the student passes all courses:

- Print the aggregate percentage (formatted to two decimal places).
- Print the grade based on the aggregate percentage.

If the student fails any course (marks < 40 in any course), print:

- "Fail".

Note:

- Aggregate Percentage: Average performance across all courses, calculated by summing all marks and dividing by the number of courses.

passorFa...

Submit

```
1 # write your code here
2 n = int(input())
3 marks = list(map(int, input().split()))
4
5 if min(marks) < 40:
6     print("Fail")
7 else:
8     avg = sum(marks) / n
9     print("Aggregate Percentage: {:.2f}".format(avg))
10
11 if avg > 75:
```

Average time0.008 s8.00 ms

Maximum time0.012 s12.00 ms

5 out of 5 shown test case(s) passed

5 out of 5 hidden test case(s) passed

Test case 1 12 ms Debug

Expected output	Actual output
4	4
55 65 78 89	55 65 78 89
Aggregate Percentage: 71.75	Aggregate Percentage: 71.75
Grade: First-Division	Grade: First-Division

Test case 2

Terminal

Test cases

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1.1.5. Multiplication Table

00:20

Write a Python program that takes an integer as input and prints the multiplication table for that integer from 1 to 10.

Input Format:

- The first line of input contains an integer that represents the number for which the multiplication table is to be printed.

Output Format:

- Print the multiplication table for the given number in the format:

<number> x <multiplier> = <result>

Note:

- The character "x" is used in the output to represent multiplication.
- Refer to the visible test-cases for more clarity.

Sample Test Cases

multiplica...

```
1 #write your code here...
2 # Read the integer input
3 n = int(input())
4
5 # Loop from 1 to 10 inclusive
6 for i in range(1, 11):
7     # Calculate the result
8     result = n * i
9     # Print using the specific format required:
10    <number> x <multiplier> = <result>
11    # Note: Using an f-string is the cleanest way to
```

Average time

0.004 s

3.50 ms

Maximum time

0.004 s

4.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 4 ms

Debug

Expected output

Actual output

8

8

8 x 1 = 8

8 x 1 = 8

8 x 2 = 16

8 x 2 = 16

8 x 3 = 24

8 x 3 = 24

8 x 4 = 32

8 x 4 = 32

8 x 5 = 40

8 x 5 = 40

Terminal

Test cases

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1. Practical 1

1.1. Practice Lab Assignment

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1.1.4. Reverse a Number

01:50

Write a Python program to reverse the digits of the number and print the reversed number.

Input Format:

- The input is a single integer.

Output Format:

- The output prints a single integer, which is the reversed number.

Sample Test Cases

reverseN...

Submit

```
1 #write your code here...
2 x = int(input())
3 sum = 0
4 while(x>0):
5     rem = x%10
6     sum = sum*10+rem
7     x = x // 10
8 print(sum)
9
```

Average time

0.004 s

3.60 ms

Maximum time

0.005 s

5.00 ms

2 out of 2 shown test case(s) passed

3 out of 3 hidden test case(s) passed

Test case 1

4 ms

Debug

Expected output

5367

7635

Actual output

5367

7635

Test case 2

3 ms

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1.1.3. Days between Two Dates

00:22

Write a Python program to calculate number of days between two dates.

Input Format:

- The first line contains the first date in the format *YYYY - MM - DD*.
- The second line contains the second date in the format *YYYY - MM - DD*.

Output Format:

- An integer representing the number of days between the given dates.

Note:

- The first date should always be considered earlier than the second date.

Sample Test Cases

noOfDay...

```
1 from datetime import date
2 from datetime import date
3
4 # input dates
5 y1, m1, d1 = map(int, input().split('-'))
6 y2, m2, d2 = map(int, input().split('-'))
7
8 # create date objects
9 date1 = date(y1, m1, d1)
10 date2 = date(y2, m2, d2)
11
```

Average time
0.006 s
5.75 ms

Maximum time
0.006 s
6.00 ms

2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 1 6 ms

Debug

Expected output

2014-07-02

2014-11-02

123

Actual output

2014-07-02

2014-11-02

123

Test case 2 6 ms

Terminal

Test cases

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1.1.2. Conditional Calculation Based on the Number of Digits 00:20

Write a Python program that accepts an integer n as input. Depending on the number of digits in n .

Constraints:
 $1 \leq n \leq 999$

Input Format:
• The input consists of a single integer n .

Output Format:
• If n is a single-digit number, print its square.
• If n is a two-digit number, print its square root (rounded to two decimal places).
• If n is a three-digit number, print its cube root (rounded to two decimal places).
• Else print "Invalid".

Sample Test Cases

condition...

1 #write your code here...
2 import math
3
4 def solve():
5 try:
6 # Read the input integer
7 line = input().strip()
8 if not line:
9 return
10 n = int(line)
11

Average time 0.007 s 6.57 ms
Maximum time 0.011 s 11.00 ms
4 out of 4 shown test case(s) passed
3 out of 3 hidden test case(s) passed

Test case 1 11 ms
Expected output 9
Actual output 9
81 81

Test case 2 5 ms

Test case 3

Terminal Test cases

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1. Practical 1

1.1. Practice Lab Assignment

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1.1.1. Calculate Momentum

00:45

Write a program that accepts the mass of an object (in kilograms) and its velocity (in meters per second), then calculates and displays the momentum of the object. The momentum p is calculated using the formula:

$$p = m \times v$$

where:

m is the mass of the object (in kilograms).

v is the velocity of the object (in meters per second).

Input Format:

- A single floating-point number representing the mass of the object in kilograms.
- A single floating-point number representing the velocity of the object in meters per second.

Output Format:

- The output will display calculated momentum with appropriate units (kgm/s) (rounded up to 2 decimal places).

Sample Test Cases

calculate...

```
1 #Type Content here...
2 mass = float(input())
3 velocity = float(input())
4 momentum = mass * velocity
5 print(f"{momentum:.2f}kgm/s")
```

Average time

0.007 s

7.00 ms

Maximum time

0.009 s

9.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 8 ms

Debug

Expected output

5.0

10.0

50.00kgm/s

Actual output

5.0

10.0

50.00kgm/s

Test case 2 6 ms

Terminal

Test cases

< Prev

Reset

Submit

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